

Solutions for Problem Set 8

Mathematics for Social Scientists

EXERCISE 1. (*Essential Mathematics for Economic Analysis (5th edition)*, Knut Sydsaeter and Peter Hammond, Section 15.3, Question 5)

Consider the three matrices $\mathbf{A} = \begin{pmatrix} 2 & 2 \\ 1 & 5 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ 3 & 2 \end{pmatrix}$ and $\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

(i) Find a matrix $\mathbf{C}$ such that $(\mathbf{A} - 2\mathbf{I})\mathbf{C} = \mathbf{I}$.

(ii) Is there a matrix $\mathbf{D}$ satisfying $(\mathbf{B} - 2\mathbf{I})\mathbf{D} = \mathbf{I}$?

(i) $\mathbf{C} = \begin{pmatrix} -\frac{3}{2} & 1 \\ \frac{1}{2} & 0 \end{pmatrix}$

(ii) No. ($\det(\mathbf{B} - 2\mathbf{I}) = 0$)

EXERCISE 2. Determine the rank of the following matrices:

(i)

$$\mathbf{A} = \begin{pmatrix} 3 & 7 \\ -7 & 3 \end{pmatrix}$$

(ii)

$$\mathbf{B} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 2 & 3 & 1 \end{pmatrix}$$

(i) $\det(\mathbf{A}) = 9 - (-49) \neq 0 \quad \Rightarrow \quad \mathbf{A} \text{ has full rank} \quad \Rightarrow \quad \text{rank}(\mathbf{A}) = 2$

(ii) $\text{rank}(\mathbf{B}) = 2$

EXERCISE 3. Compute the determinant of the following matrices. If they are invertible, find

the inverse.

$$C = \begin{pmatrix} 2 & 7 \\ 7 & 8 \end{pmatrix}, \quad D = \begin{pmatrix} x & x^2 \\ x^3 & x^4 \end{pmatrix}, \quad E = \begin{pmatrix} -3 & 0 & 4 \\ 1 & 4 & -2 \\ 7 & -6 & 3 \end{pmatrix}.$$

- $\det C = -33, C^{-1} = \begin{pmatrix} -8/33 & 7/33 \\ 7/33 & -2/33 \end{pmatrix}$
- $\det D = 0$
- $\det E = -136, E^{-1} = \begin{pmatrix} 0 & 3/17 & 2/17 \\ 1/8 & 37/136 & 1/68 \\ 1/4 & 9/68 & 3/34 \end{pmatrix}$

EXERCISE 4. Does the property $\det(A+B) = \det(A) + \det(B)$ hold for $A, B \in \mathbb{R}^{n \times n}$ and any $n \geq 1$?

No. You should provide a counterexample. Here is one: $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$. In the exam, you should show with a counterexample that the property doesn't hold.

EXERCISE 5. Compute the determinant of the following matrices:

$$A = \begin{pmatrix} 2 & -3 & 2 \\ 1 & 4 & 0 \\ 0 & 1 & -5 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 & 2 \\ 4 & 0 & 3 \\ 3 & 2 & 1 \end{pmatrix}.$$

Are they invertible?

- $\det A = -53$
- $\det B = 21$
- Both are invertible

EXERCISE 6. Find eigenvalues and eigenvectors of the matrix

$$B = \begin{pmatrix} 7 & 3 \\ 3 & -1 \end{pmatrix}.$$

- $\lambda_1 = 8, \lambda_2 = -2$
- For λ_1 , $L = \left\{ \begin{pmatrix} 3a \\ a \end{pmatrix} : a \in \mathbb{R} \right\}$
for λ_2 , $L = \left\{ \begin{pmatrix} a \\ -3a \end{pmatrix} : a \in \mathbb{R} \right\}$

EXERCISE 7. Compute the characteristic polynomial $P_C(\lambda)$ for $C = \begin{pmatrix} 0 & -1 & 1 \\ -3 & -2 & 3 \\ -2 & -2 & 3 \end{pmatrix}$ and try to find the eigenvalues!

Characteristic polynomial: $(\lambda - 1)(\lambda + 1)(\lambda - 1)$;

Eigenvalues: $\lambda_1 = \lambda_3 = 1$; $\lambda_2 = -1$

EXERCISE 8. Does the matrix $D = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ have (real-valued) eigenvalues?

No. Characteristic polyn.: $\lambda^2 + 1 = 0 \Rightarrow \lambda_1 = i; \lambda_2 = -i$, where $i = \sqrt{-1}$